

3

C

Distance travelled during 0.20 m s^{-2} acceleration = $\frac{20^2}{2(0.20)} = 1000 \text{ m}$

Distance travelled for the 0.40 m s^{-2} deceleration = $\frac{20^2}{2(0.40)} = 500 \text{ m}$

Total time travel = $\frac{20}{(0.20)} + \frac{1500}{20} + \frac{20}{(0.40)} = 225 \text{ s}$

4

D

Given that the reading is smaller than the person's mass, this means that the